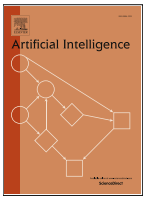
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HiMAE: High-intensity mask-based autoencoder framework for unsupervised human action recognition

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ABSTRACT

Reducing dependence on labeled data through self-supervised learning is a central objective in Artificial Intelligence, and this paradigm has been widely investigated in the context of skeleton-based action recognition. Notably, masked prediction-based techniques have greatly improved pre-training performance. However, most of these methods fail to explicitly exploit contextual dependencies during the masking process despite the crucial role this information plays in accurate action recognition. In this paper, we argue that explicitly modeling human contextual dependencies through comprehensive masked representation reconstruction is more effective for unsupervised human motion recognition. Based on such an idea, we propose a High-intensity Mask-based Autoencoder (HiMAE) for unsupervised 3D action recognition. In HiMAE, we develop two kinds of high-intensity masks to identify features within action sequences that exhibit significant variations and uncertainty. In this way, the proposed HiMAE could explicitly capture human contextual dependencies at both spatial and temporal levels. Specifically, the High-Entropy Component Extraction Block is proposed to learn underlying topological correlations in the spatial context via leveraging a high-entropy-making strategy. Extensive experimental NTU-60, NTU-120, and PKU-MMD datasets show that HiMAE pre-training greatly improves performance with vanilla transformers, achieving state-of-the-art results without bells and whistles.

1. Introduction

Due to its numerous applications in human-robot interaction, 3D skeleton-based human action recognition has attracted a lot of attention and has been thoroughly explored over the past decade [1–4]. By utilizing a variety of architectures, such as Transformer-based networks [5], Recurrent Neural Networks (RNN) [6,7], and Graph Convolutional Networks (GCN) [8–10], Supervised Skeletons-based action recognition techniques have demonstrated impressive performance. Since these models are fully-supervised during training, a substantial quantity of labeled data is required. Skeletal sequence annotation is a laborious and time-consuming process. Furthermore, under supervision, overfitting can occur, particularly in transformer models with large capacities and mild inductive bias. A central goal of Artificial Intelligence is to learn transferable representations from raw data with minimal labels. Consequently, the challenge of learning discriminative representations from unlabeled skeleton poses, known as self-supervised skeleton-based action recognition, is becoming a critical area of interest.

All self-supervised techniques fall into one of two groups depending on learning paradigms: reconstruction-based or contrastive learning-based. With their clear structure and encouraging results, contrastive learning-based techniques have demonstrated incred-

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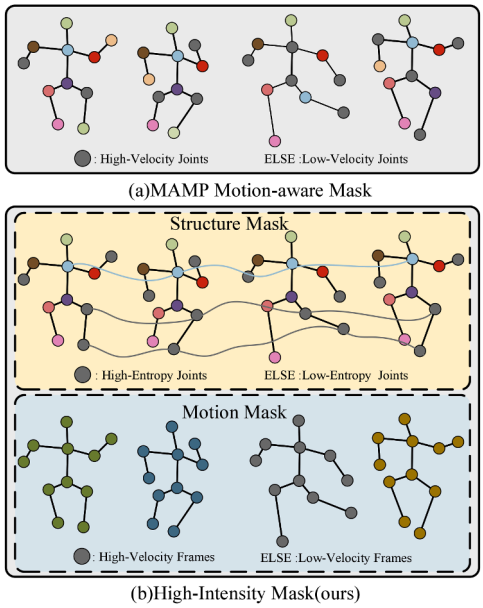


Fig. 1. A comparative illustration of the mask strategy between Motion-aware masking and our High-Intensity masking. High-Intensity utilizes entropy in spatial dimensions to guide spatial masking and velocity in temporal dimensions to guide temporal masking, instead of utilizing velocity as a masking strategy in both temporal and spatial dimensions.

ible promise [11–17] . Skeletal transformation is used in these techniques to produce positive and negative samples. For instance, Li et al. [11] presented a single/cross-view contrastive learning framework to capture comprehensive representation. Thoker et al. [12] applied inter-skeleton contrast learning to enrich the skeleton sparse input, leading to high-level semantics capture. Yet, it still has certain drawbacks as a global representation learner, which prevent it from exploring 3D actions further. These include the over-reliance on heuristic action data augmentations and the absence of explicit restrictions for temporal context modeling. Therefore, many attempts focus on reconstruction-based methods [18–23] to obtain effective motion dependencies by predicting skeleton data. e.g., Su et al. [18] proposed an encoder-decoder recurrent neural network to reconstruct skeleton sequences in a re-generation manner.

Among these reconstruction-based methods, MAE-based methods [19,20,22–26] show a promising performance via reconstructing skeleton sequences from the masked action features. These masked features are obtained by specific mask auto-encoder architecture. Consequently, the model can be directed to learn compact features from these masked representations during the self-reconstruction process. e.g., [19] designed global and local attention mechanisms to learn the natural joint dynamics across frames via an attention mask. Yan et al. [23] developed the SkeletonMAE structure, leveraging prior human topology knowledge to guide the reconstruction of the masked skeleton joints. However, most methods did not explicitly exploit contextual dependencies during the masking process, which carry crucial information for action recognition. A recent work [22] designed a motion-aware masking strategy, which utilizes motion intensity as a prior semantic to guiding reconstructing temporal motion, while the model only considers modeling temporal-wise contextual dependencies (see Fig. 1(a)). Another work [20] adopted a frame and joint-making strategy for spatial-temporal contextual dependencies modeling; however, such a spatial-temporal mask strategy is based on a random mask with a specific mask ratio, which did not explicitly exploit contextual dependencies. For instance, the action of "sit down" tends to be recognized based on those high-entropy joints that exhibit significant variations and uncertainty, e.g., hip, neck, rather than ankle and knee. This requires the model to learn rich and effective feature representations from those high-entropy joints. However, the random mask in joints can not ensure effective feature modeling from these high-entropy joints.

Motivated by this observation, we develop two kinds of high-intensity masks: high-entropy masks, masking joints that exhibit significant variations and uncertainty in the spatial-wise context, and high-velocity masks, masking frames that contain high-velocity information in the temporal-wise context (see Fig. 1(b)). These two high-intensity masks attempt to identify features within action sequences that exhibit significant variations and uncertainty, as these features often carry crucial action-related information. Thus, by exploring these feature representations, the model can be guided to learn richer and more effective feature representations, thereby enhancing the performance and accuracy of self-supervised 3D skeleton recognition tasks.

Based on these two kinds of high-intensity masks, we propose a High-intensity Mask-based Autoencoder (HiMAE) using the high-intensity mask to guide the reconstruction process at both spatial and temporal levels. The central idea of HiMAE is to make masking itself semantically purposeful. Existing skeleton MAE methods mainly use masking to remove part of the input and then reconstruct the missing skeleton sequence or motion signal. However, different masking choices impose different pretext tasks. Random or ratio-based masking encourages generic reconstruction, whereas motion-aware masking emphasizes dynamic temporal regions. HiMAE

further asks which spatial and temporal components should be hidden so that their reconstruction requires contextual reasoning. We therefore use entropy as a spatial prior to mask joints with high topological uncertainty, and velocity as a temporal prior to mask frames with strong motion dynamics. This transforms masked autoencoding from generic missing-token recovery into high-intensity contextual dependency learning. In our HiMAE, we build a parallel structure (see Fig. 2), which consists of two key components: Spatial-wise Topological Contextual Correlations Learning (STCL), learning underlying topological correlations across skeletal space, and Temporal-wise Motion Contextual dependencies Learning (TMCL), guiding HiMAE pays more attention to semantically rich temporal regions. In STCL, we develop a High-Entropy Component Extraction Block (HE-CEB) to capture topological contextual correlations via a high-entropy mask. While a High-Velocity Component Extraction Block (HV-CEB) is proposed to classify motion contextual dependencies via a high-velocity mask.

Extensive experiments are conducted on three public datasets, i.e., NTU RGB+D 60 [27] and NTU RGB+D 120 [28]. We demonstrate that the proposed HiMAE could significantly outperform the state-of-the-art approaches under various comparison metrics. The main contributions are summarized as follows:

- We propose the High-intensity Mask-based Autoencoder (HiMAE) to pay effective attention to semantically rich temporal and spatial regions, thus resulting in more comprehensive and efficient feature representation learning and achieving effective motion recognition.
- In our HiMAE, the High-Entropy Component Extraction Block (HE-CEB) is developed to capture topological contextual correlations, which is leveraged as prior semantic richness to guide the model for compact and rich representation learning.
- Extensive experiments are conducted on three public benchmarks to verify the effectiveness of our method. Under semi-supervised evaluation, HiMAE outperforms MAMP [22] by 11.8% and 12.8% on the NTU RGB+D 60 XSub and NTU RGB+D 60 XView settings.

2. Related work

2.1. Skeleton-based action recognition

In computer vision, skeleton-based action recognition is a basic yet difficult job. While geometric relationships between skeleton joints were often the focus of traditional approaches [29–31], deep neural networks have been used in recent advances to address this task [6,32–35]. To process body keypoints better, Du et al. [6] established a hierarchical RNN. Additionally, attention-based techniques have been created to automatically choose video frames and significant skeleton joints for adaptive learning [32–34, 36]. Nevertheless, gradient vanishing is a common issue for RNNs, which poses optimization difficulties [37]. The ability of Graph Convolution Networks (GCNs) to extract temporal and spatial information from skeletal data has drawn attention recently. In order to improve the flexibility of graphical representations, attention mechanisms have been included by capturing discriminative features based on spatial composition and temporal dynamics [9,34,38] , as suggested by Yan et al. [39] in their proposal of spatial-temporal GCNs. Recent work has also explored frequency-aware self-supervised skeleton representation, such as FAGAR [17], which uses frequency-domain augmentation and adaptive frequency compression to improve robustness to dynamic spatio-temporal variations.

2.2. Reconstruction-based learning

Reconstructing masked skeletons for long-term global motion dynamics was initially suggested by Zheng et al. [40]. By forecasting masked skeleton data, the subsequent approaches [18–23,40] are able to capture the spatial-temporal correlation. The primary areas of study for researchers are reconstruction targets and masking tactics. PREDICT & CLUSTER [18] proposes a pioneering system that associates input sequences of body key points with specific actions without the need for labeled data. GL-Transformer [19] introduces a global-local motion transformer, which enhances action representation by capturing both global and local motion patterns. SSL [23] proposes a graph-based masked autoencoder for pre-training skeleton sequences, leveraging graph structures to capture temporal dependencies and improve the quality of action representations. MAMP [22] introduces masked motion predictors, demonstrating their efficacy as strong learners for 3D action representation by leveraging masking mechanisms to improve feature quality. However, the model only considers modeling temporal-wise contextual dependencies. In parallel, SkeletonMAE [20] adopted a frame and joint-making strategy for spatial-temporal contextual dependencies modeling; however, such a spatial-temporal mask strategy is based on a random mask with a specific mask ratio, which did not explicitly exploit contextual dependencies.

More recent studies have further moved from random or purely motion-based masking toward adaptive target selection. STJD-MP [24] introduces spatio-temporal joint density to identify prime joints by measuring the interaction between moving and static joints, and assigns higher masking probabilities to joints with larger density variations. ACA2Net [26] explores context-guided masked modeling through an asymmetric teacher-student framework, where motion-aware and motion-compass-aware masking are combined with adaptive alignment to transfer richer contextual representations. ActCLR+ [25] incorporates semantic-aware masked motion modeling into actionlet-dependent contrastive learning, showing that reconstructing informative local motion regions can improve skeleton representation learning.

Different from these methods, we develop a High-intensity Mask-based Autoencoder, which is less about adding another masking score and more about redefining the pretext task: the model is trained not merely to reconstruct what is missing, but to infer the most context-demanding spatial and temporal components from the remaining skeleton sequence.

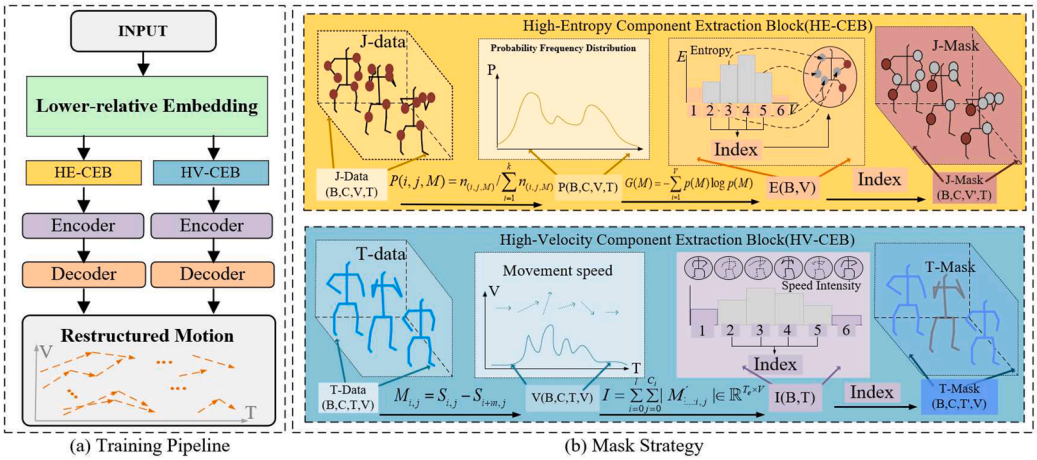


Fig. 2. The proposed HiMAE framework's total pipeline. We employ a spatio-temporal dual-stream network for modeling. In the Spatial-wise Topological Contextual Correlations Learning (STCL), we utilize the high-entropy component masking strategy to guide the masking process. In the Temporal-wise Motion Contextual dependencies Learning (TMCL), we employ the High-Velocity Component masking strategy to guide the masking process. Additionally, we use a spatio-temporal separation masking strategy to allow the model to focus more on semantically rich regions separately in spatial and temporal dimensions.

3. Method

3.1. Overview

The overall framework of the proposed HiMAE is illustrated in Fig. 2. Given the input skeleton sequences, these are fed into both the Spatial-wise Topological Contextual Correlations Learning (STCL) and the Temporal-wise Motion Contextual Dependencies Learning (TMCL). Specifically, it takes skeleton sequences $x_t \in \mathbb{R}^{T_s \times V \times C_s}$ and $x_j \in \mathbb{R}^{V \times T_s \times C_s}$ as input, which are extended to a consistent temporal length T_s by randomly selecting **subsequences** (i.e., short temporal clips) from the original skeleton data. In this case, C_s stands for the coordinate channels, while V indicates the number of joints. Additionally, the motion sequence $M \in \mathbb{R}^{T_s \times V \times C_s}$ is derived from the input by computing the temporal differential (with manual padding for the first frame).

At STCL, the input $x_j \in \mathbb{R}^{V \times T_s \times C_s}$ is mapped into Lower-relative Embedding $E_j \in \mathbb{R}^{V \times T_s \times C_d}$. while $x_t \in \mathbb{R}^{T_s \times V \times C_s}$, as input, is mapped into Lower-relative Embedding $E_t \in \mathbb{R}^{T_s \times V \times C_d}$ at TMCL. Next, most of the highest entropy embedding features are hidden by applying the High-Entropy Component Extraction Block (HE-CEB) to E_j. Similarly, most of the greatest velocity embedding characteristics are hidden when the High-Velocity Component Extraction Block (HV-CEB) is applied to E_t. The unmasked features are processed using an encoder-decoder framework: the transformer encoder employs the learnable masked tokens and the latent representation from the transformer encoder for context modeling, and the transformer decoder extracts representations from the unmasked joint embedding. Reconstructing the motion information from the input skeleton sequence is our ultimate objective. Only the transformer encoder and the Lower-relative Embedding layer are retained for tasks that come after the pre-training procedure is finished.

3.2. Spatial-wise topological contextual correlations learning (STCL)

The functionality of Spatial-wise Topological Contextual Correlations Learning (STCL) is to extract topological contextual correlations indicating considerable uncertainty or diversity across skeletal space, which guides the model toward compact and effective topological correlation learning. For example, the relative positional relationships between certain skeletal nodes may vary significantly across different actions, or the connectivity patterns between nodes may change. Given the input $x_j \in R^{V \times T_s \times C_s}$, it is linearly mapped into Lower-relative Embedding $E_j \in \mathbb{R}^{V \times T_s \times C_d}$. Afterward, E_j is masked by employing High-Entropy Component Extraction Block (HE-CEB), effectively concealing most of the highest entropy embedding features. The unmasked features are handled using the Motion Reconstruction, the Transformer encoder captures representations from the unmasked joint embedding, while the transformer decoder uses the learnable masked tokens and the latent representation from the transformer encoder for context modeling.

3.2.1. Lower-relative embedding

In earlier transformer-based methods, each spatio-temporal skeleton joint is embedded individually, which incurs significant computational overhead. To mitigate these costs, our approach fuses the input skeleton sequence $x_j \in R^{V \times T_s \times C_s}$ into non-overlapping fragments in the time dimension. $x'_j \in R^{V \times T_e \times l \times C_s}$,where l represents the length of each segment and $T_e = T_s/l$. In each segment, the joints containing the information of the entire segment are embedded together. It is worth noting that we use CNN based on the time

dimension to implement JointEmbed:

$$E = JointEmbed(x'_j) \in R^{V \times T_e \times C_d}, \tag{1}$$

where C_d denotes the dimensionality of these features. When juxtaposed with the original skeleton data, the temporal granularity of the resultant embedding E is diminished by a multiple of l. leading to a more efficient computational process.

3.2.2. High-entropy component extraction block (HE-CEB)

The spatial dimension of human skeletal sequences differs from the temporal dimension. Between adjacent frames in the temporal dimension, there is a clear correspondence, and the velocity information between adjacent frames con-tains rich semantic information. However, evaluating the importance of skeletal points using velocity information [22] in the spatial dimension is inappropriate. In spatial topology, the most semantically informative joints are those whose coordinates vary in class-discriminative ways across time (e.g., hands/feet in "throw", hip/neck in "sit down"). Entropy summarizes how spread/uncertain a joint's distribution is across the sequence; masking these high-entropy joints forces the model to reconstruct them from the remaining body graph and thus to learn topological contextual correlations rather than memorizing trivial cues. We, thus, leverage entropy to assess the semantic depth of skeletal points within the spatial dimension.

To guide the embedding feature masking in HE-CEB, we use the skeleton point entropy as the empirical semantic richness prior. We extract the joint sequence $J^{joint} \in R^{V \times T_e \times C_s}$,where T_e is the number of frames in the temporal dimension and V is the number of skeleton joints. Then, the joint entropy ,which indicates the joint significance of each joints ,is computed as follows. We first calculate the probability frequency distribution of the spatial dimension of the human skeletal sequence:

$$\begin{aligned} f_{(i,j,M)} &= \frac{n_{(i,j,M)}}{T_e \cdot \Delta x}, \quad i \in \{1, \dots, V\}, \quad j \in \{1, \dots, k\}, \quad M \in \{X, Y, Z\}, \\ P(i, j, M) &= \frac{f_{i,j,M}}{\sum_{j=1}^{k} f_{i,j,M}}, \quad P(i, j, M) \in \mathbb{R}^{V \times k \times C_S}. \end{aligned} \tag{2}$$

In the above formulation, i represents the spatial dimension, and j denotes the discrete intervals along that dimension. The function $n(i, j, M)$ counts the number of samples that fall into interval j in the spatial dimension i. Here, the intervals correspond to uniform bins along the coordinate axis M ($M \in \{X, Y, Z\}$), with each bin having a width of Δx. T_e represents the total number of samples, which is also the number of frames in the temporal dimension. The term $f_{(i,j,M)}$ denotes the frequency of samples falling into interval j in spatial dimension i, normalized by $T_e \cdot \Delta x$. Finally, $P(i, j, M)$ represents the probability distribution of the coordinate values of the skeletal joint J^{joint} across the temporal dimension.

The underlying assumption is that skeleton points with higher probability frequencies contain more semantic information. This is because their consistent presence across time frames suggests they are more significant in understanding the overall motion or activity being analyzed. Entropy is used to measure the amount of information or uncertainty in the probability distribution of each skeleton point. For each joint i and coordinate axis M, the entropy is calculated over the k bins using Eq. (3). The three coordinate-wise entropies are then averaged to obtain one entropy score G_i for each joint, which is used to guide spatial masking.

$$\begin{aligned} G_i(M) &= -\sum_{j=1}^{k} P(i, j, M) \log_2 P(i, j, M), \quad i = 1, \dots, V, \quad M \in \{X, Y, Z\}, \\ G_i &= \frac{1}{3} \sum_{M \in \{X,Y,Z\}} G_i(M), \quad i = 1, \dots, V. \end{aligned} \tag{3}$$

Subsequently, we transform the joint entropy vector $\mathbf{G} = [G_1, \dots, G_V]$ into a probabilistic distribution with a temperature hyper-parameter τ, which signifies the likelihood of each skeletal joint's embedded feature being concealed. The Gumbel-Max trick is utilized to facilitate an efficient sampling process guided by probabilities:

$$\begin{aligned} q_i &= \text{Softmax}_i(\mathbf{G}/\tau), \quad i = 1, \dots, V, \\ g_i &= -\log(-\log(\sigma_i)), \quad \sigma_i \in U[0, 1], \\ idx_{\text{joint}}^{\text{mask}} &= \text{Top-K-Index}_i(\log q_i + g_i), \end{aligned} \tag{4}$$

where q_i is the masking probability of joint i, and the obtained idx_{joint}^{mask} indicates which joints are masked and are adapted for the selection of unmasked tokens as detailed in Section 3.2.3. These procedures prompt the network to focus more intently on areas that are semantically rich, which in turn boosts its capacity to acquire more distinctive 3D action representations. More implementation details can be found in Algorithm 1.

3.2.3. Motion reconstruction

Inspired by [22], we leverage an encoder-decoder architecture for motion reconstruction (see Fig. 3). Concretely, the transformer encoder primarily focuses on learning representations, while the decoder is tasked with implementing the pre-training pretext.

Encoder: In the encoder, we first augment the input joint embeddings E with separate spatio-temporal positional embeddings E_e^s and E_e^t. This is achieved by element-wise addition (with broadcasting), as follows:

$$E_p = E \oplus E_e^s \oplus E_e^t, \tag{5}$$

Algorithm 1 High-intensity masking strategy in HiMAE training.

Input: Skeleton sequence $x_j \in \mathbb{R}^{V\times T_s\times C_s}$ (spatial), $x_t \in \mathbb{R}^{T_s\times V\times C_s}$ (temporal), masking ratio $\rho = 80\%$
Output: Masked indices $idx^{\text{mask}}_{\text{joint}}$ (spatial), $idx^{\text{mask}}_{\text{frame}}$ (temporal)

Spatial Masking (HE-CEB)

Embed x_j into $E_j \in \mathbb{R}^{V\times T_e\times C_d}$ using JointEmbed with segment length l
Extract joint sequence $\boldsymbol{J}^{\text{joint}} \in \mathbb{R}^{V\times T_e\times C_s}$
for each coordinate axis $M \in \{X, Y, Z\}$ **do**
    Compute frequency $f_{(i,j,M)} = \frac{\eta_{(i,j,M)}}{T_e\cdot\Delta x}$ for $k = 10$ bins $j = 1, \ldots, k$
    Normalize to probability $P(i, j, M) = \frac{f_{(i,j,M)}}{\sum_{j=1}^{k} f_{(i,j,M)}}$
    Calculate joint entropy $G_i(M) = -\sum_{j=1}^{k} P(i, j, M)\log_2 P(i, j, M)$ for $i = 1, \ldots, V$
end for
Aggregate joint entropy $G_i = \frac{1}{3}\sum_{M\in\{X,Y,Z\}} G_i(M)$
Apply Softmax with temperature $\tau = 0.80$: $q = \text{Softmax}(G/\tau)$
Sample with Gumbel-Max: $g = -\log(-\log(\sigma))$, $\sigma \sim U[0, 1]^V$
Select top-ρ indices: $idx^{\text{mask}}_{\text{joint}} = \text{Top-K-Index}(\log q + g)$

Temporal Masking (HV-CEB)

Embed x_t into $E_t \in \mathbb{R}^{T_s\times V\times C_d}$ using JointEmbed
Compute motion $M \in \mathbb{R}^{T_s\times V\times C_s}$: $M_{i,j,k} = x_{i,j,k} - x_{i-t,j,k}$, stride $t = l$
Pad M with replication: $M_{0:t-1,j,k} = M_{1:2t-1,j,k}$
Segment M into $M' \in \mathbb{R}^{T_e\times V\times l\times C_s}$
Compute motion intensity $I = \sum_{i=0}^{V}\sum_{j=0}^{l}\sum_{k=0}^{C_s} |M'_{i,i,j,k}|$
Apply Softmax with temperature τ: $P = \text{Softmax}(I/\tau)$
Sample with Gumbel-Max: $g = -\log(-\log(\sigma))$, $\sigma \sim U[0, 1]^{T_e}$
Select top-ρ indices: $idx^{\text{mask}}_{\text{frame}} = \text{Top-K-Index}(\log P + g)$
Return: $idx^{\text{mask}}_{\text{joint}}$, $idx^{\text{mask}}_{\text{frame}}$

where $\oplus$ denotes element-wise addition. After retrieving the indices from the HE-CEB, we select the unmasked tokens from E_p. These tokens are then flattened to produce E^u_p. Next, the latent representation is obtained through a series of standard Transformer blocks:

$$\begin{aligned} H_0 &= E^u_p, \\ H'_l &= \text{MSA}(\text{LN}(H_{l-1})) + H_{l-1}, \quad l \in 1, \ldots, L_e \\ H_l &= \text{FFN}(\text{LN}(H'_l)) + H'_l, \quad l \in 1, \ldots, L_e \\ H^u_e &= \text{LN}(H_{L_e}). \end{aligned} \tag{6}$$

Each Transformer block consists of a **Multi-head Self-Attention (MSA)** module followed by a **Feed-Forward Network (FFN)**. Residual connections are applied at each stage, followed by **Layer Normalization (LN)** to stabilize training and enhance feature learning. Here, H_{L_e} represents the final latent feature representation after L_e Transformer layers, which encodes the essential information from the unmasked token embeddings.

Decoder: In the decoder, maskable tokens are integrated into H^u_e in accordance with the mask indices idx^{mask}. The subsequent embeddings are then restructured to H_e. As with the encoder, spatial positional embeddings $E^{s\prime}_e$ and temporal positional embeddings $E^{t\prime}_e$ are appended to each token, assisting the masked tokens in ascertaining their respective positions.

Motion Prediction: Our approach targets the motion sequence M^{target}, normalized by segment-wise mean and standard deviation, rather than rebuilding the original skeletons. We use a prediction head to anticipate the temporal motion of masked human joints given the decoded feature Z_d:

$$M^{\text{pred}}_j = \text{MotionPred}(Z_d). \tag{7}$$

Our findings indicate that a straightforward fully connected layer provides adequate outcomes for this task. For the joints of humans that are masked, we calculate the mean squared error (MSE) between the forecasted outcome M^{pred}_j and the target for reconstruction M^{target}.

$$\begin{aligned} Mask_j &= \text{Stacking}(idx^{mask}_{joint}) \in \mathbb{R}^{T_e\times V\times(l\times C_z)}, \\ \mathcal{L}_1 &= \frac{1}{|Mask_j|}\sum Mask_j \times \|(M^{\text{pred}}_{i,j,:} - M^{\text{target}}_{i,j,:})\|^2_2, \end{aligned} \tag{8}$$

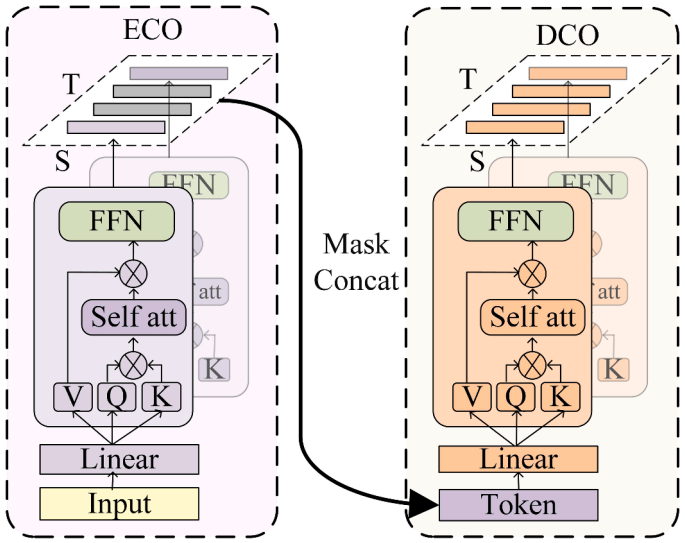


Fig. 3. The structure of Encoder(EC) and Decoder(DC).

where Repeat means replicate the tensor. This revised presentation aims to convey the essential information while reducing the similarity with the original text.

3.3. Temporal-wise motion contextual dependencies learning (TMCL)

The functionality of Temporal-wise Motion Contextual Dependencies Learning (TMCL) is to classify motion contextual dependencies exhibiting significant changes or dynamics over time sequences, which could guide models toward more comprehensive and efficient motion representations. For instance, the positional or velocity changes of a skeletal node across different time steps. Such variations often carry crucial action-related information, hence are considered rich in information content. Inspired by Mao et al. [22], we leverage High-Velocity Component Extraction Block (HV-CEB) to capture temporal-wise motion contextual dependencies. Given $x_t \in R^{T_s \times V \times C_s}$, it as input is linearly mapped into Lower-relative Embedding $E_t \in \mathbb{R}^{T_s \times V \times C_d}$. After that, E_t is masked by employing a Motion Mask via a High-Velocity Component Extraction Block, effectively concealing the majority of the highest-velocity embedding features. The unmasked features are handled using the Motion Reconstruction, the transformer encoder captures representations from the unmasked joint embedding, while the transformer decoder uses the learnable masked tokens and the latent representation from the transformer encoder for context modeling.

3.3.1. High-velocity component extraction block (HV-CEB)

Human skeleton sequences have a significant amount of semantic information, with clear correlations between consecutive frames. By utilizing temporal differencing on the joint coordinates, we can effectively capture motion data. Let $S \in \mathbb{R}^{T_s \times V \times C_s}$ denote the preprocessed skeleton sequence, where $T_s = 120$ is the number of frames, V is the number of joints, and $C_s = 3$ represents the X, Y, Z coordinates. Thus, $S_{i,:,:} \in \mathbb{R}^{V \times C_s}$ is the complete skeleton pose at frame i. The motion data, represented as $M \in \mathbb{R}^{T_s \times V \times C_s}$, is computed by temporal differencing:

$$M_{i,:,:} = S_{i,:,:} - S_{i-t,:,:}, \qquad i = t, t+1, \ldots, T_s - 1, \tag{9}$$

where t is the stride that determines the step size for capturing motion. To facilitate processing, the motion sequence is padded to match the length of the original input sequence as follows:

$$M_{0:t-1,:,:} = \begin{cases} 0, & \text{if padded with constants} \\ M_{t:2t-1,:,:}, & \text{if padded by replication} \end{cases} \tag{10}$$

In this context, padding can be done by either filling the initial t frames with zeros (constant padding) or by replicating the first t frames of the motion sequence (replicate padding).

Motion information has two functions in our methodology: first, it serves as an objective for reconstruction in the pre-training phase, and second, it provides an empirical measure of semantic richness that directs the masking of embedding characteristics. We extract the motion information $M \in R^{T_i \times V \times C_i}$ with stride $t = l$ when the skeleton sequence is embedded in segments of length l. We then apply duplicate padding in accordance with Eqs. (9) and (10). Then based on the segmentation operation in the Section 3.2.1, this motion sequence is molded to $M' \in R^{T_e \times V \times l \times C_i}$. The significance of motion in each temporal segment is represented by the motion

intensity I, which is then calculated as follows:

$$I = \sum_{i=0}^{V} \sum_{j=0}^{l} \sum_{k=0}^{C_i} |M'_{:,i,j,k}| \in \mathbb{R}^{T_e}. \tag{11}$$

The $idx_{\text{frame}}^{\text{mask}}$ is obtained according to the Eqs. (3) and (4). These indicates which temporal segment are masked and are adapted for the selection of unmasked tokens as detailed in Section 3.3.2.

3.3.2. Motion reconstruction

The encoder-decoder architecture is according to the Section 3.2.3. Noted that, in Motion Prediction, Replicate means replicate the tensor element-wise.

$$\begin{aligned} Mask_t &= \text{Padding}(idx_{frame}^{mask}) \in \mathbb{R}^{T_e \times V \times (l \times C_z)}, \\ \mathcal{L}_2 &= \frac{1}{|Mask_t|} \sum Mask_t \times \|(M_{i,j,:}^{\text{pred}} - M_{i,j,:}^{\text{target}})\|_2^2. \end{aligned} \tag{12}$$

Our method's ultimate loss function is the sum of L_1 and L_2:

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2. \tag{13}$$

4. Experiments

4.1. Datasets

The NTU RGB+D 60 dataset [27]: NTU-60 is a large collection of human action data captured using three Microsoft Kinect v2 sensors. It includes 56,578 samples covering 60 action categories performed by 40 subjects aged 10 to 35. Each sample contains information about 25 key joints and can involve up to two subjects. The dataset is commonly divided into two benchmarks: Cross-Subject (CS) and Cross-View (CV). CS splits training and validation data based on different subjects, while CV splits based on sensor views. For semi-supervised learning, researchers use varying proportions of labeled data (5, 10, 20, or 40). Overall, it's a valuable resource for skeleton-based action recognition research, offering diverse action categories and subjects, with benchmarks for cross-subject and cross-view evaluations, and support for semi-supervised learning.

The NTU RGB+D 120 dataset [28]: NTU-120 is an expansion of the NTU RGB+D dataset, offering a larger-scale collection for human action recognition. It includes 113,945 samples covering 120 action classes performed by 106 subjects. Two benchmarks are defined for evaluation: Cross-Subject (CS) and Cross-Setup (CE). In the CS benchmark, there are 63,026 training samples and 50,919 testing samples from 53 subjects. In the CE benchmark, data is divided based on setup IDs, with 54,468 sequences from even IDs for training and 59,477 sequences from odd IDs for testing. Notably, for the pretrain + linear evaluation protocol, the entire labeled data is used for training.

The PKU-MMD dataset [41]: For skeletal action recognition tasks, PKU-MMD is divided into two phases: PKU-MMD I and PKU-MMD II. With 20,000 and 7000 action examples, respectively, covering 51 categories and single or two performers, it is a challenging task because of its wide range of categories and little amount of training data. In accordance with the cross-subject methodology, we separate action occurrences into training and test sets after cropping them out based on temporal annotations, as per [22] .

4.2. Implementation details

Data Preprocessing: We employed the data preprocessing method from Mao et al. [22], which entails uniform sampling from the skeleton sequence. Initially, a consecutive segment is randomly extracted from the original skeleton sequence with a ratio p (p is randomly chosen from the range [0.5,1] during pre-training and is set to 0.9 for downstream tasks). This cropped segment is subsequently scaled to a constant length T_s via bilinear interpolation, with T_s typically being set to 120.

Entropy-guided masking settings: After interpolation, we set $T_e = T_s = 120$ for entropy estimation. For each coordinate axis $M \in \{X, Y, Z\}$, the sample-specific coordinate range is divided into $k = 10$ equal-width bins shared by all joints in that sample. The bin probabilities are computed from the temporal frequencies, and the base-2 Shannon entropy is calculated over the bin index. The entropy values from the three coordinate axes are averaged to obtain one score per joint. We use a fixed temperature $\tau = 0.80$ in $\text{Softmax}(G/\tau)$, without temperature annealing, and apply an 80% masking ratio. The masked joints are selected with the Gumbel-Max trick; zero-probability bins use the convention $0 \log_2 0 = 0$ and the Gumbel sampling uses a numerical stability constant of 10^{-10}. These settings are fixed across all datasets.

Network Architecture: In our proposed HiMAE, we adopt a vanilla transformers as the backbone network, consisting of $L = 8$ identical building blocks. Each structural unit is equipped with an embedding dimension of 256, an 8-head multi-head self-attention module, and a feed-forward network with a hidden layer dimension of 1024. Inputs are processed with learnable spatio-temporal positional embeddings that are applied separately. The transformer decoder used for pre-training mirrors the configuration of the backbone encoder, with the exception that the number of layers L_d is reduced to 3. The workflow of HiMAE can be found in Algorithm 2.

Pre-training Details: Eighty percent of the input embedding tokens are masked during the pre-training phase. The target motion data is enhanced with zero-padding and uses a stride of $m = 1$. The network is pre-trained with a batch size of 256 across 400 epochs.

Table 1
Comparison of performance using NTU-60, NTU-120, and PKU-MMD datasets in a linear assessment process. Methods are classified into Contrastive Learning-based and Mask-based categories. The top-performing outcome is bolded, while the second-best outcome is underlined.

Method	Input Stream	NTU-60		NTU-120		PKU-MMD	
		X-sub	X-view	X-sub	X-set	Phase I	Phase II
Contrastive Learning-based Methods							
CrosSCLR [13]	Joint + Motion + Bone	77.8	83.4	67.9	66.7	84.9	21.1
3s-AimCLR [11]	Joint + Motion + Bone	78.9	83.8	68.2	68.8	–	–
3s-C-F [42]	Joint + Motion + Bone	79.1	87.2	69.2	70.8	89.2	49.8
3s-HYSP [43]	Joint + Motion + Bone	79.1	85.2	64.5	67.3	–	–
3s-ActCLR [44]	Joint + Motion + Bone	84.3	88.8	74.3	75.7	90.0	–
CMD [14]	Joint Only	84.1	90.9	74.7	76.1	–	52.6
Hico-Transformer [15]	Joint Only	81.1	88.6	72.8	74.1	89.3	49.4
RMMD [45]	Joint Only	83.0	90.5	75.2	75.8	90.3	50.6
STJD-CL [24]	Joint Only	82.3	87.9	70.5	72.8	90.6	51.5
FAGAR [17]	Joint Only	80.8	85.8	71.3	72.1	90.4	–
ActCLR + [25]	Joint Only	82.3	88.2	70.9	73.2	–	–
Mask-based Methods							
SkeAttnCLR [46]	Joint + Motion + Bone	82.0	86.5	77.1	80.0	89.5	**55.5**
MAMP [22]	Joint Only	84.9	89.1	78.6	79.1	**92.2**	<u>53.8</u>
STJD-MP [24]	Joint Only	85.4	90.2	79.1	80.4	–	–
ACA^2Net [26]	Joint Only	**86.0**	89.6	79.1	79.8	<u>92.0</u>	53.7
Ours	Joint Only	**86.0**	**91.3**	**80.3**	**82.8**	91.1	52.8

Table 2
Comparing the semi-supervised protocol's performance using the NTU 60 datasets. The ultimate performance was calculated by averaging the outcomes of five runs.

Method	NTU-60			
	X-sub		X-view	
	(1%)	(10%)	(1%)	(10%)
LongT GAN [40]	35.2	62.0	–	–
MS2L [47]	33.1	65.1	–	–
3s-Cros-SCLR [13]	51.1	74.4	50.0	77.8
3s-Colorization [42]	48.3	71.7	52.5	78.9
3s-Hi-TRS [48]	49.3	77.7	51.5	81.1
3s-AimCLR [11]	54.8	78.2	54.3	81.6
3s-CMD [14]	55.6	79.0	55.5	82.4
3s-HYSP [43]	–	80.5	–	85.4
RMMD [45]	58.3	77.1	56.0	81.0
3s-SkeAttnCLR [46]	59.6	81.5	59.2	83.8
MAMP [22]	66.0	88.0	68.7	91.5
3s-STJD-CL [24]	67.5	82.5	68.4	88.0
ActCLR + [25]	70.5	82.2	73.1	86.0
ACA^2Net [26]	66.2	88.9	69.4	91.9
Ours	**77.8**	**90.0**	**81.5**	**94.4**

Using the AdamW optimizer, we set the betas to (0.9, 0.95) and apply a weight decay factor of 0.05. Starting at 0, the learning rate increases linearly to 1e-3 in the first twenty warm-up epochs, then decreases to 5e-4 in accordance with a cosine decay schedule. The PyTorch framework is utilized to conduct the tests, utilizing the computing power of NVIDIA RTX 4090 GPUs.

Linear evaluation protocol: The pre-trained encoder's parameters are maintained constant in the context of linear evaluation in order to extract features, and a trainable linear classifier is then used for the classification job. Using a batch size of 256 and a momentum factor of 0.9, the SGD (Stochastic Gradient Descent) optimizer trains the model across 100 epochs. The learning rate starts at 0.1 and is gradually decreased to 0 by adjustments made in accordance with a cosine decay plan.

Semi-finetune evaluation protocol: Just 1% and 10% of the training dataset are used for fine-tuning in the semi-supervised assessment setup, in line with the consistency needed in other training circumstances.

Transfer Learning Evaluation : In the transfer learning evaluation process, The network is first pre-trained on a source dataset and then fine-tuned on a different target dataset in the transfer learning assessment phase. This method is used to determine how flexible and broadly applicable the learnt representations are. NTU-60 and NTU-120 are used as the source datasets, while the PKU-MMD II dataset is used as the target for fine-tuning.

Table 3
Comparing performance in the protocol of transfer learning. PKU-II is the target dataset, whereas NTU-60 and NTU-120 are the source datasets.

Method	To PKU-II	
	NTU-60	NTU-120
LongT GAN [40]	44.8	–
MS2L [47]	45.8	–
ISC [12]	51.1	52.3
CMD [14]	56.0	57.0
RMMD [45]	53.7	55.0
MAMP [22]	70.6	73.2
ActCLR+ [25]	62.1	–
ACA^2Net [26]	**73.6**	73.3
Ours	71.5	**73.8**

Algorithm 2 Training and testing workflow of HiMAE.

Input: Skeleton sequence dataset $D = \{x_i\}_{i=1}^{N}$, where $x_i \in \mathbb{R}^{T \times V \times C_s}$
Output: Trained HiMAE model and evaluation metrics

Training Phase

Initialize HiMAE model with vanilla Transformer backbone ($L = 8$, embedding dim = 256)
Set hyperparameters: batch size = 256, epochs = 400, masking ratio = 80%
for each epoch $e = 1$ to 400 **do**
    Sample mini-batch $B = \{x_j\}_{j=1}^{256}$ from D
    Preprocess B: random crop with ratio $p \in [0.5, 1]$, interpolate to $T_s = 120$
    Compute motion sequence M via temporal differencing with padding
    Generate high-intensity masks (see Algorithm 1)
    Forward pass through STCL and TMCL streams:
        Lower-relative embedding: $E_j \leftarrow \text{JointEmbed}(x_j)$, $E_t \leftarrow \text{JointEmbed}(x_t)$
        Apply HE-CEB on E_j and HV-CEB on E_t
        Encode unmasked tokens, decode with masked tokens
    Compute loss: $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$ (MSE on masked joints and frames)
    Update model parameters using AdamW with cosine learning rate schedule
end for

Testing Phase (Linear Evaluation)

Freeze pre-trained encoder parameters
Add linear classifier on end of encoder
Train classifier with SGD (batch size = 256, epochs = 100, momentum = 0.9)
for each test sample x_{test} in D_{test} **do**
    Preprocess x_{test}: crop with $p = 0.9$, interpolate to $T_s = 120$
    Extract features using pre-trained encoder
    Predict action label using linear classifier
end for
Compute accuracy on dataset

4.3. Experimental result and analysis

Linear: As can be seen from Table 1, HiMAE, which utilizes the masked prediction strategy, shows considerable benefits when juxtaposed with contrastive learning techniques. Additionally, HiMAE surpasses other methods that employ masked prediction and contrastive learning on the majority of the datasets. Notably, on the NTU-60 XView and NTU-120 XSet datasets, HiMAE achieves a performance improvement of 2.2% and 3.7%, respectively, over the previously leading method MAMP. Furthermore, our approach yields results that are on par with the current state-of-the-art across the PKU-MMD datasets.

Semi-supervised evaluation: The HiMAE model we've introduced consistently outperforms previous methodologies that are founded on the masked prediction task. This includes methods such as MAMP [22] and SkeAttnCLR [46], and this outperformance

Table 4
Comparison of model complexity, runtime, and accuracy on NTU-60 XSub. Runtime is averaged over multiple runs on an NVIDIA RTX 4090 with a batch size of 128.

Method	Complexity (M)	Runtime (ms)	Accuracy (%)
CMD [14]	135.37	**0.516**	84.1
MAMP [22]	**6.35**	3.21	84.9
Ours	12.7	5.65	**86.0**

Table 5
Ablation study on the model structure. The results are reported on the NTU-60 XSub and NTU-120 XSub datasets under the linear protocol.

Structure	NTU-60	NTU-120
Only SMD	82.9	77.7
Only TMR	82.9	77.8
SMD + TMR	**86.0**	**80.3**

is observed across all datasets (refer to Table 2). Furthermore, our approach exhibits particularly competitive performance on the NTU-60 XView and NTU-120 XSet datasets.

Transfer Learning Evaluation: Using NTU-60 and NTU-120 as source datasets, respectively, the results shown in Table 3 demonstrate that HiMAE outperforms the state-of-the-art technique MAMP by 0.9% and 0.6%. This demonstrates the robustness and strength of the qualities that the HiMAE framework has taught us.

Complexity and Runtime: To further evaluate the practical efficiency of HiMAE, we conducted an analysis of model complexity and inference runtime, comparing it against two competitive baselines: CMD [14] and MAMP [22]. All experiments were performed on an NVIDIA RTX 4090 GPU with a batch size of 128, and the reported runtimes represent the average inference time per sample over multiple runs to ensure reliability. Table 4 summarizes these results alongside the top-1 accuracy on the NTU-60 XSub dataset. CMD exhibits the highest complexity at 135.37 million parameters (M) but achieves the fastest runtime of 0.516 ms per sample, reflecting its efficiency in processing, though its accuracy (84.1%) is the lowest among the three. MAMP, with a lightweight design of 6.35 M parameters, requires 3.21 ms per sample and achieves an accuracy of 84.9%. In contrast, HiMAE, with a moderate complexity of 12.7 M parameters, takes 5.65 ms per sample but delivers the highest accuracy of 86.0%. This trade-off indicates that HiMAE's enhanced performance comes at the cost of increased computational overhead, primarily due to its dual-stream architecture (STCL and TMCL) and high-intensity masking strategy, which enrich feature representations but extend processing time. These results highlight HiMAE's suitability for applications prioritizing accuracy over real-time constraints, while CMD and MAMP may be preferred in latency-sensitive scenarios.

4.4. Ablation study

Effects of STCL and TMCL: In the proposed HiMAE, we build a parallel structure including Spatial-wise Topological Contextual Correlations Learning (STCL) and Temporal-wise Motion Contextual Dependencies Learning (TMCL). To verify the superiority of STCL and TMCL, we further investigate several variants of our HiMAE by 1). Only SMD-STCL, 2). Only TMR-TMCL, 3) HiMAE (STCL + TMCL). From Table 5, it can be observed that deploying both STCL and TMCL can achieve the best performance in action recognition. In contrast, using only STCL or only TMCL results in a similar decline in performance. We attribute this to the lack of comprehensive and effective masked representation reconstruction, which weakens the network's ability to capture rich implicit relationships.

Effects of High-intensity Masking Strategy: In the proposed HiMAE, we develop high-intensity masks consisting of high-entropy masks and high-velocity masks to identify features within action sequences that exhibit significant variations and uncertainty. To explore the effects of high-intensity Masking Strategy applied at both spatial and temporal levels, we further investigate several variants of our high-intensity masking Strategy by:

- using High-Entropy masks at both Spatial and temporal levels (noted as HE2HE).
- using High-Velocity masks at both spatial and temporal levels (noted as HV2HV).
- using High-Entropy masks at temporal levels and High-Velocity masks at spatial levels, respectively (noted as HE2HV).
- using High-Velocity masks at temporal levels and High-Entropy masks at spatial levels, respectively (noted as HV2HE).

Fig. 5 shows the accuracy results on NTU-CS-60. It can observed that recognition performance suffered from the deployment of High-Velocity masks at spatial levels (HV2HV, HE2HV) when compared with the deployment of High-Entropy masks at spatial levels (HV2HE, HE2HE). This suggests that the deploying of High-Entropy masks at spatial levels, which is responsible for masking joints that exhibit significant variations and uncertainty in the spatial-wise context, contributes to guiding the model to learn the rich

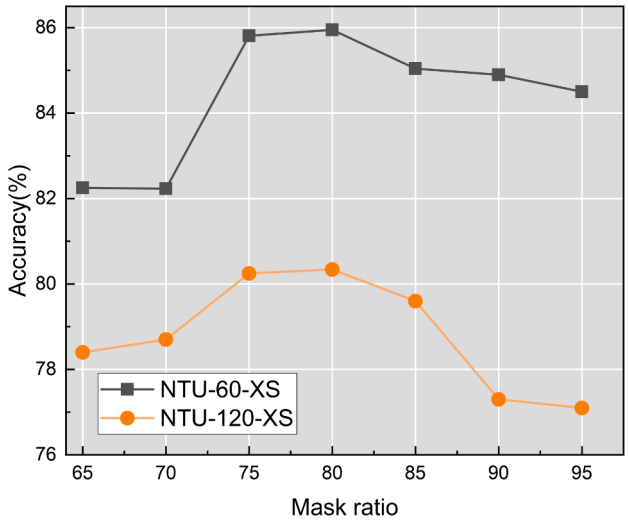


Fig. 4. Ablation study on high-intensity masking ratio for recognition performance on NTU-60 and NTU-120 X-sub under the linear protocol, respectively.

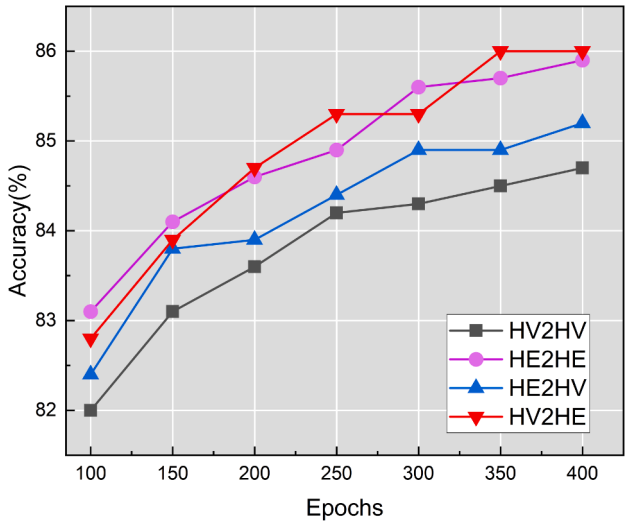


Fig. 5. Ablation study on high-intensity masking strategy. The results are reported on the NTU-60 under linear protocol.

topological contextual correlations. The effectiveness of high-intensity masking strategy is illustrated when deploying HV2HE results in noticeable performance deterioration when compared with HE2HE.

Effects of Masking Ratio: To explore the optimal high-intensity masking ratio on action modeling, we test the model performance with various making ratios on NTU-60 and NTU-120 X-sub, respectively (see Fig. 4). It is evident that either too large or too small masking ratios have a negative impact on recognition performance. The model can achieve the optimal results with a masking ratio of 80%.

4.5. Qualitative analysis

4.5.1. t-SNE visualization

To qualitatively assess the effects of the masking ratio on learned features, we executed a comparative analysis between different masking ratios by visually depicting the feature embeddings. Fig. 6 samples 15 action classes from the NTU-60 and NTU-120 dataset by t-SNE algorithm[64]. It can be observed that the feature embeddings learned by deploying a masking ratio of 80% are more compactly clustered than other masking ratios, which demonstrates the superiority of our proposed high-intensity mask.

To further compare the discriminability of learned representations with an existing masked-motion method, Fig. 7 visualizes the t-SNE embeddings of HiMAE and MAMP on NTU-60 and NTU-120. On NTU-60, both methods produce recognizable class groups, while HiMAE preserves more compact class-wise structures for several neighboring clusters. The difference is more apparent on

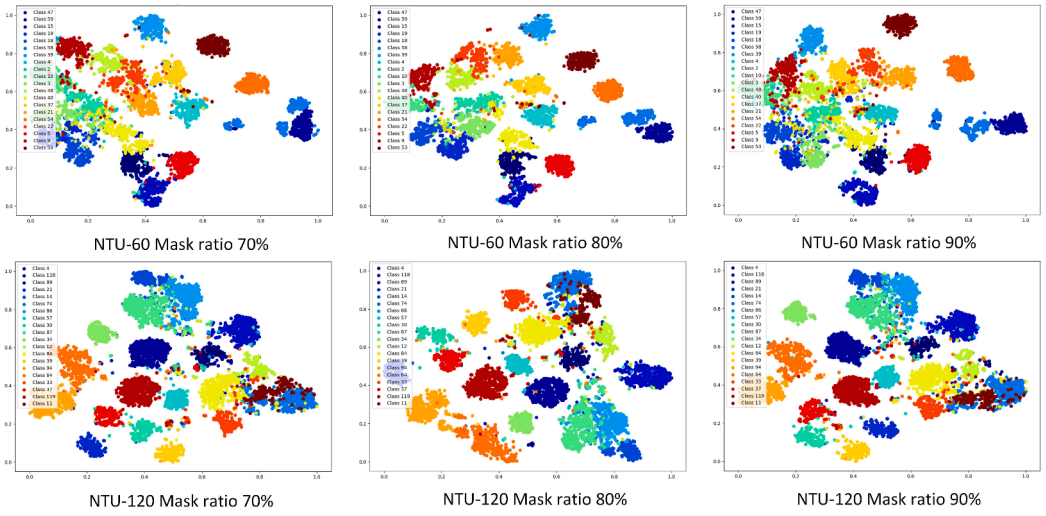


Fig. 6. t-SNE visualization of learned features with different mask ratios on NTU-60 and NTU-120, respectively.

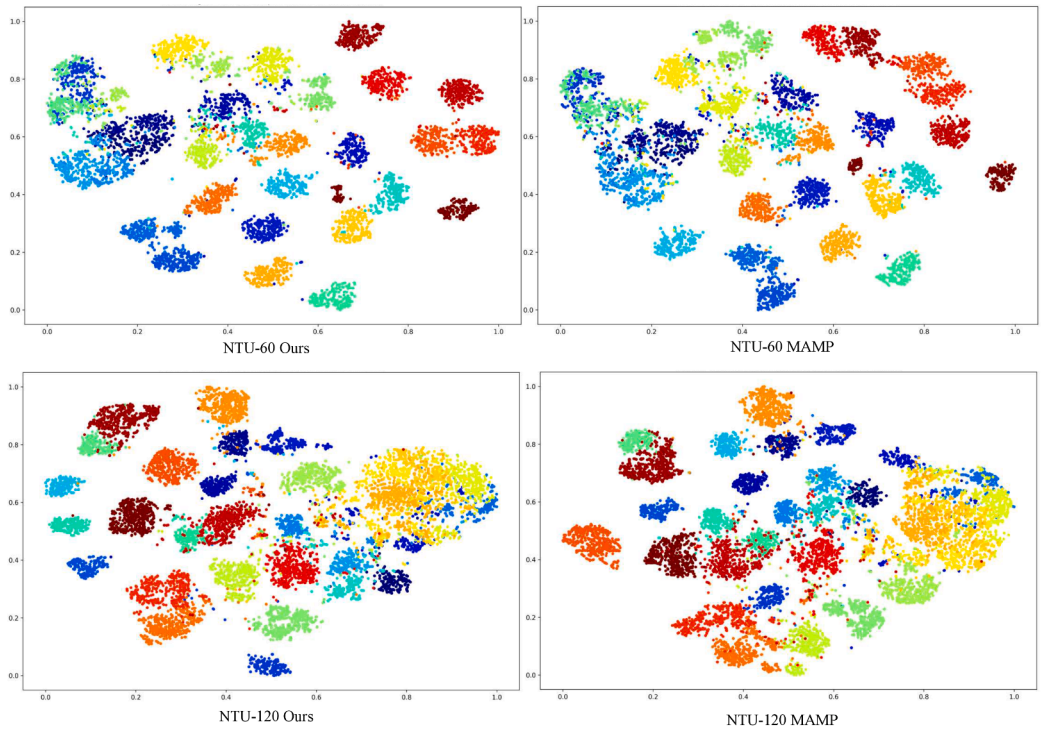


Fig. 7. t-SNE visualization comparing representations learned by HiMAE (left) and MAMP (right) on NTU-60 (top) and NTU-120 (bottom). Each color denotes one action class. HiMAE yields more compact intra-class clusters and reduced inter-class overlap, particularly on NTU-120.

the more challenging NTU-120 dataset: HiMAE features exhibit tighter intra-class aggregation and clearer boundaries between most classes, whereas the MAMP embedding contains more dispersed clusters and substantial mixing among neighboring classes. These observations are consistent with the quantitative results in Table 1 and indicate that HiMAE learns a representation with improved intra-class consistency and inter-class separability. This comparison supports the motivation for high-intensity masking: the complementary high-velocity temporal and high-entropy spatial masks require reconstruction of context-demanding motion transitions and joint configurations, reducing reliance on generic pose cues and improving action discrimination.

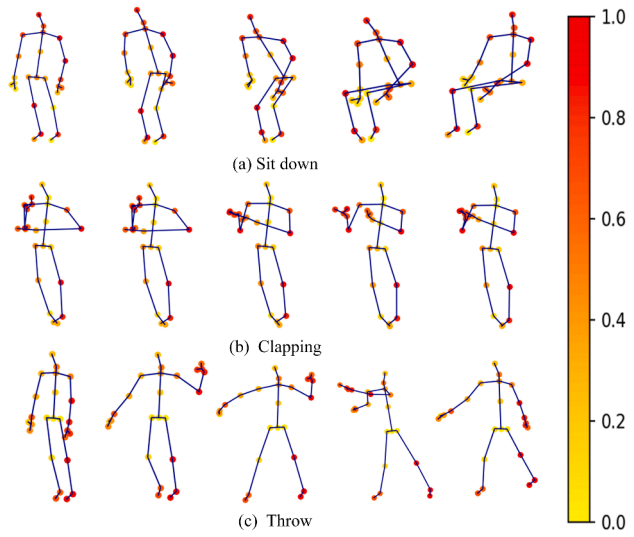


Fig. 8. Visualization of sample-specific high-entropy joints selected by the high-entropy masking strategy. From top to bottom, we illustrate (a) 'Sit down,' (b) 'Clapping,' and (c) 'Throw,' respectively. Note that red joints indicate relatively high coordinate-distribution entropy within the displayed sequence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.5.2. High-entropy masks visualization

In HiMAE, the high-entropy mask is designed to identify spatial components whose coordinate distributions exhibit larger temporal uncertainty. To illustrate this behavior, we visualized the skeleton joint distribution on the NTU-60 dataset in Fig. 8. Note that the highlighted joints are sample-specific high-entropy joints computed from the temporal distribution of joint coordinates, rather than manually defined semantic key joints or class-level average saliency maps. A high entropy value indicates that the coordinate distribution of a joint is more dispersed across the sequence, suggesting stronger contextual reconstruction demand. Therefore, joints involved in posture transition, balance adjustment, or compensatory coordination may also be selected, even if they are not the most visually dominant action part.

For example, in "sit down", the selected ankle joints may reflect foot placement and balance adjustment during the standing-to-sitting transition. In "clapping", the highlighted knee and ankle may correspond to lower-body stabilization accompanying repetitive upper-limb movement. In "throw", the hip can remain relatively stable as the body pivot, while the limbs exhibit larger temporal coordinate variation. These observations suggest that HE-CEB encourages the model to reconstruct spatial components that require contextual reasoning over the whole body structure, rather than simply focusing on visually dominant joints.

4.5.3. Qualitative comparison of confusing actions

Fig. 9 compares "point to something with finger" and "take a selfie", two actions with a similar upright body posture but different local upper-limb dynamics. "point to something with finger" is characterized by a rapid arm extension and the directional hand-elbow-shoulder configuration away from the torso. In contrast, "take a selfie" requires lifting and maintaining the hand near the head, so the discriminative evidence lies in the arm trajectory and the local hand-head relationship. The motion-intensity and entropy maps show that these cues are concentrated in a small number of fast-moving frames and in specific upper-limb joints, rather than being uniformly distributed over the sequence or skeleton.

This visualization is consistent with the class-wise linear-evaluation results. HiMAE obtains 94% accuracy for "point to something with finger" and 76% for "taking a selfie", compared with 21% and 42%, respectively, for the model trained with random masking. With temporal masking alone, randomly removing a segment that contains a high-velocity arm transition can eliminate the very evidence needed to distinguish these actions. The spatial high-entropy mask complements temporal masking: it preserves the temporal evolution while hiding selected joints, requiring the model to infer the informative arm-torso and arm-head configurations from the visible body structure. Consequently, high-velocity temporal masking learns when discriminative motion occurs, whereas high-entropy spatial masking learns which joints and body-relative relations produce it. Their combination promotes reconstruction of the context-demanding components shown in Fig. 9, yielding a substantially more discriminative representation than random masking for these two actions.

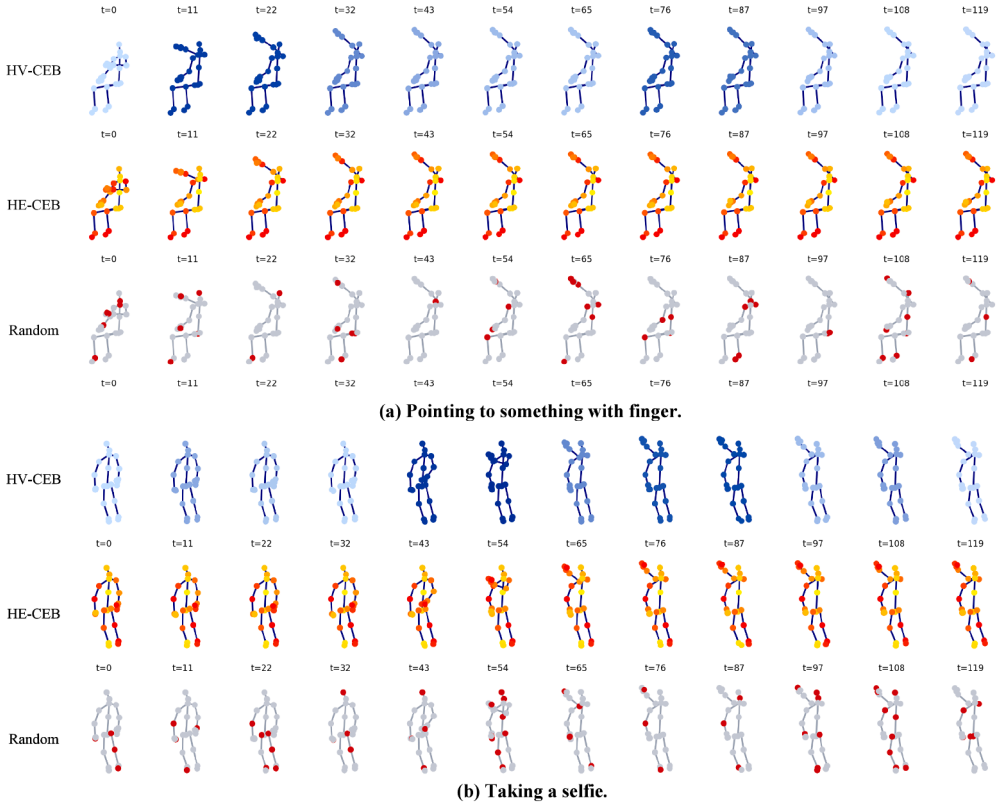


Fig. 9. Qualitative visualization of two visually similar NTU-60 actions: "point to something with finger" (top three rows) and "take a selfie" (bottom three rows). Each column shows one of 12 uniformly sampled frames. For each action, the first row visualizes joint motion intensity (darker blue indicates higher intensity), the second row visualizes joint-coordinate entropy (warmer colors indicate higher entropy), and the third row highlights the high-entropy joints selected by HE-CEB (red). The high-intensity components concentrate on the discriminative upper-limb motion and its body-relative configuration. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Conclusion and future work

This paper addresses unsupervised human motion recognition from the view of mask-based autoencoder. In contrast to previous studies that emphasize masked joint self-reconstruction, we insist that leveraging explicit human contextual dependencies through comprehensive masked representation reconstruction is more effective for unsupervised human motion recognition. Building on this insight, we introduce a High-intensity Mask-based Autoencoder (HiMAE), which employs a high-intensity mask to guide the reconstruction process at both spatial and temporal levels. This approach significantly enhances the modeling capability of contextual dependencies. Specifically, we develop the High-Entropy Component Extraction Block (HE-CEB) to learn underlying topological correlations in the spatial context. Experimentally, our HiMAE shows great generality and performance on three public datasets. Detailed derivations and ablations are provided to testify the effectiveness of each component in the proposed HiMAE.

To enhance the focus on critical joints and movements, future work could involve refining dynamic masking strategies within the HiMAE framework. Dynamic masking entails adjusting the intensity and pattern of masks in real-time, based on the complexity and variability of input motion sequences. This adjustment can be achieved through adaptive algorithms, allowing the model to prioritize regions of interest that demonstrate significant motion or semantic importance, thereby improving the contextual understanding of human actions.

Addressing the challenges observed in the PKUMMD II dataset, where performance might have been hindered due to inadequate attention to meaningful data in the temporal dimension, integrating more sophisticated techniques beyond the High-Velocity Component Extraction Block (HV-CEB) masking strategy could be beneficial. By exploring alternative approaches or refining the existing masking strategies, the model could better focus on capturing the nuanced information present in the data, potentially leading to improved performance on datasets like PKUMMD II.

CRediT authorship contribution statement

Jianqi Zhong: Validation, Methodology, Conceptualization; **Zhenrong Wang:** Writing – original draft, Methodology, Data curation; **Xu Tan:** Supervision, Project administration; **Wenming Cao:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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